

EXPERIMENT 7

DATA VISUALIZATION USING TABLEAU

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Sample Dataset:

Artist Streaming Dataset

Algorithm 1:

Connecting Data Source

Step 1 Open Tableau Desktop

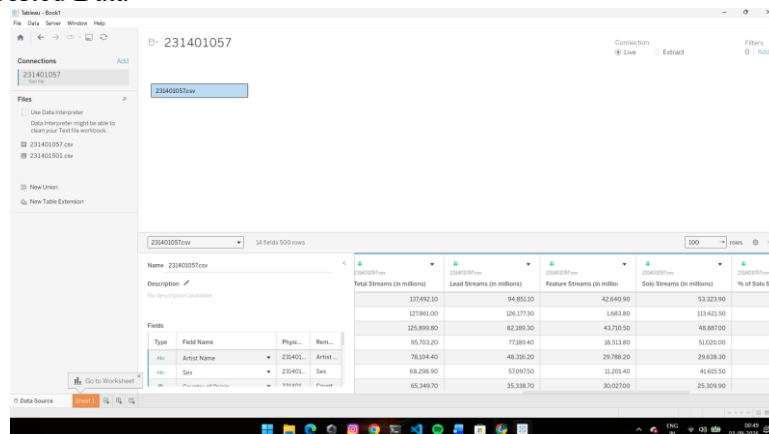
Step 2 Select Connect

Step 3 Choose Text File

Step 4 Browse Dataset

Step 5 Open Dataset

Step 6 Verify Imported Data



Name	Description	Fields	Total Streams (in millions)	Lead Streams (in millions)	Feature Streams (in million)	Solo Streams (in millions)	% of Solo Stream
231401057.csv	No description available.		127,602.00	94,350.30	42,640.90	53,323.90	
			127,601.00	126,177.30	1,663.80	133,421.90	
			125,899.80	82,389.30	43,750.50	48,867.00	
			95,703.20	77,389.40	38,513.80	51,000.00	
			78,004.40	48,338.20	29,788.20	29,638.30	
			68,298.90	57,087.50	11,201.40	41,605.50	
			65,349.70	35,338.70	30,027.00	25,309.90	

Algorithm 2:

Creating Worksheet

Step 1 Open Sheet 1

Step 2 Drag Primary Genre to Columns

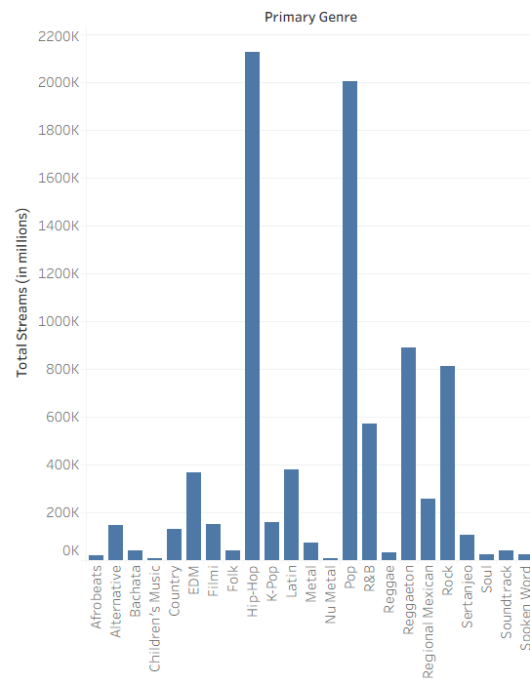
Step 3 Drag Total Streams (in millions) to Rows

Step 4 Select Visualization

Step 5 Analyze Results

Output:

Sheet 1



Algorithm 3:

Bar Chart Creation

Step 1 Drag Primary Genre → Columns

Step 2 Drag Total Streams (in millions) → Rows

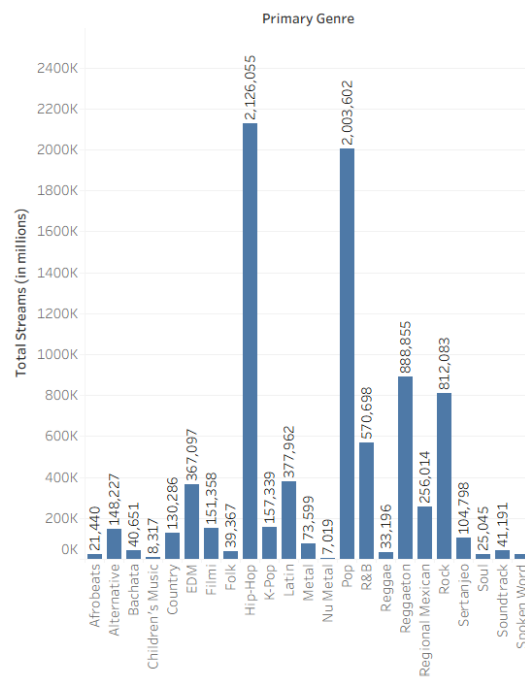
Step 3 Select Bar Chart

Step 4 Apply Labels

Step 5 Format Visualization

Output:

Sheet 1



Algorithm 4:

Pie Chart Creation

Step 1 Drag Primary Genre

Step 2 Drag Total Streams (in millions)

Step 3 Select Pie Chart

Step 4 Show Labels

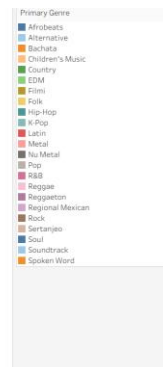
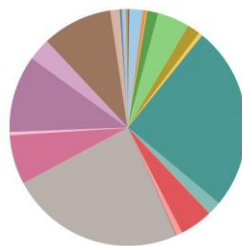
Step 5 Analyze Distribution

Output:

Genre-wise Stream Contribution

Contribution of genres to total streams.

Genre-wise Stream Contribution



Hip-Hop contributes the highest share of total streams, approximately 25.28%.

Algorithm 5:

Line Chart Creation

Step 1 Drag Artist Name

Step 2 Drag Total Streams (in millions)

Step 3 Select Line Chart

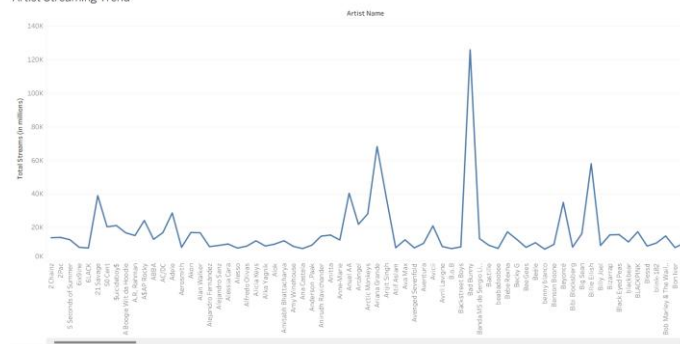
Step 4 Apply Formatting

Output:

Artist Streaming Trend

Streams trend among artists.

Artist Streaming Trend



Algorithm 6:

Creating Calculated Field

Objective

Calculate Percentage

Streams Formula (Total Streams / Total Streams) * 100

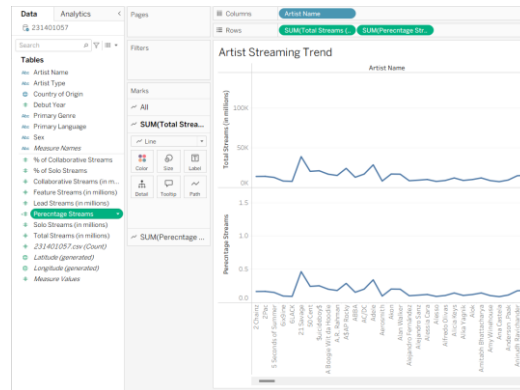
Procedure

Analysis → Create Calculated Field

Name: Percentage Streams

Formula: [Total Streams (in millions)] / { FIXED : SUM([Total Streams (in millions)]) } * 100

Output:



Algorithm 7:

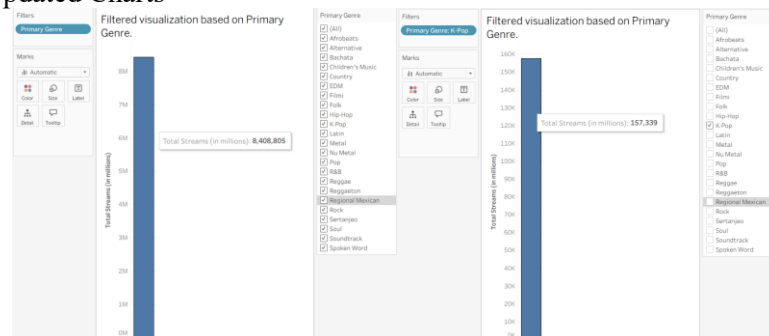
Applying Filters

Step 1 Drag Primary Genre to Filters

Step 2 Select Primary Genre

Step 3 Apply Filter

Step 4 Observe Updated Charts



Algorithm 8:

Creating Parameter

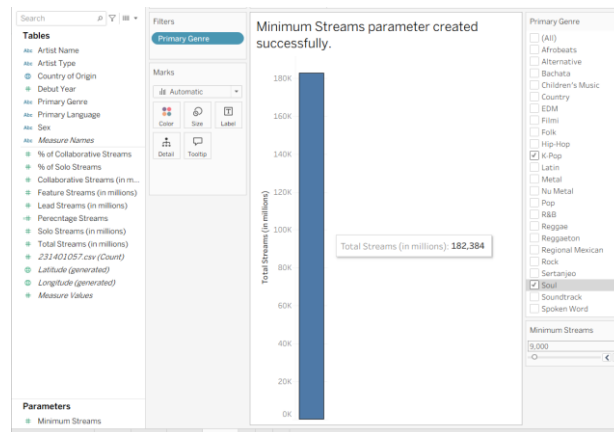
Step 1 Create Parameter

Step 2 Specify Minimum Streams

Step 3 Apply Parameter

Step 4 Display Matching Records

Output:



Algorithm 9:

Dashboard Creation

Step 1 Create Multiple Worksheets

Step 2 Select Dashboard

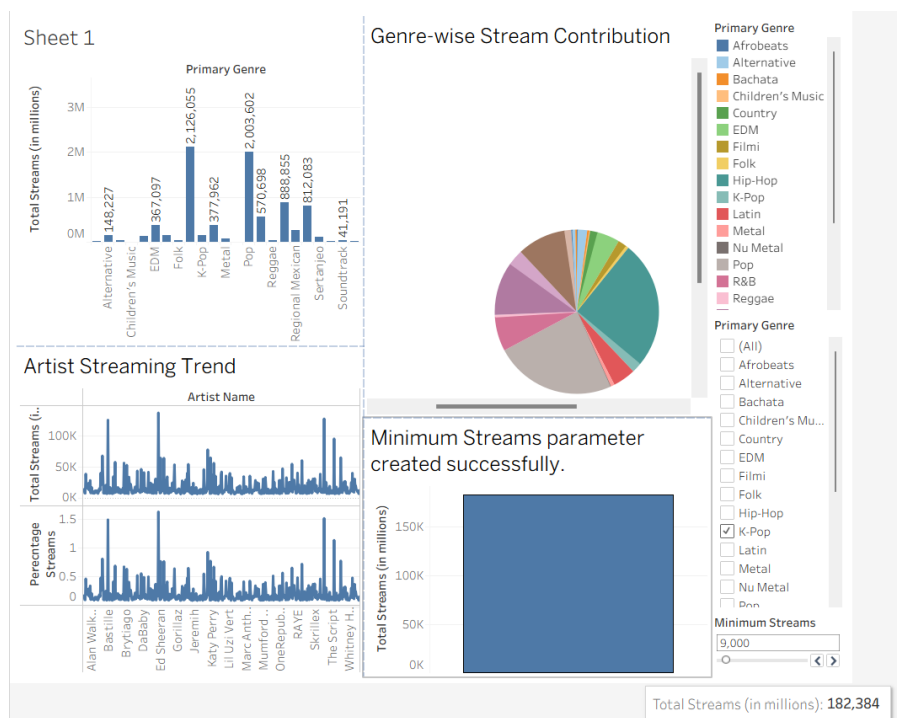
Step 3 Drag Worksheets

Step 4 Arrange Components

Step 5 Add Filters

Step 6 Save Dashboard

Output:



Result:

Thus, Tableau was used successfully to connect the artist streaming dataset, create worksheets, apply filters and parameters, generate visualizations, and build dashboards for data analysis.